In the closed Loop ABEFA.

$$3I_1 + 2(I_1 + I_2) = 35$$
$$3I_1 + 2I_1 + 2I_2 = 35$$
$$5I_1 + 2I_2 = 35 \quad (1)$$

In the closed loop BCDEB

$$4I_2 + 2(I_1 + I_2) = 40$$
$$4I_2 + 2I_1 + 2I_2 = 40$$
$$2I_1 + 6I_2 = 40 \quad (2)$$
$$(1) \times 3, \quad 15I_1 + 6I_2 = 105 \quad (3)$$
$$2I_1 + 6I_2 = 40$$
$$(3) - (2), \quad 13I_1 = 65$$
$$I_1 = \frac{65}{13} = 5A.$$

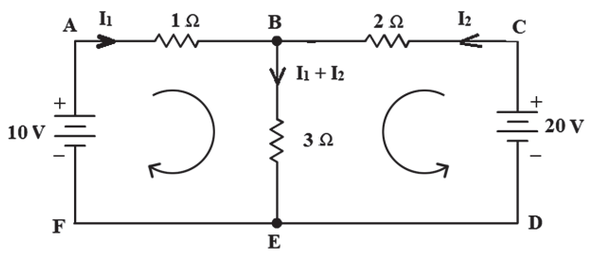
Substituting the value of $I_1$ in equation 2, we get

$$2I_1 + 6I_2 = 40 \quad (2)$$
$$2 \times 5 + 6I_2 = 40$$
$$10 + 6I_2 = 40$$
$$6I_2 = 40 - 10 = 30$$
$$6I_2 = 30$$
$$I_2 \frac{30}{6} = 5A$$

Then current in 2Ω resistor is $I_1 + I_2$ = 5 + 5 = 10A

**Example: 2**

Find the current in the 3Ω resistor in the circuit as shown.




In the closed Loop ABEFA

$$I_1 + 3(I_1 + I_2) = 10V$$
$$I_1 + 3I_1 + 3I_2 = 10 \quad (1)$$
$$4I_1 + 3I_2 = 10$$

In the closed Loop BCDEB

$$2I_2 + 3(I_1 + I_2) = 20V$$
$$2I_2 + 3I_1 + 3I_2 = 20 \quad (2)$$
$$3I_1 + 5I_2 = 20$$

$$(1) \times 3 \Rightarrow 12I_1 + 9I_2 = 30 \text{ ------- } (3)$$
$$(2) \times 4 \Rightarrow 12I_1 + 20I_2 = 80$$
$$(3) - (4) \Rightarrow -11I_2 = -50 \quad (4)$$
$$I_2 = \frac{-50}{-11} = 4.545A = 4.55A$$

Substituting the value of $I_2$ is equal to

$$3I_1 + 5I_2 = 20$$
$$3I_1 + 5 \times 4.55 = 20$$
$$3I_1 = 20 - 22.75$$
$$= -2.75$$
$$I_1 = -\frac{2.75}{3}$$
$$= -0.916A$$
$$I_1 = -0.916A$$

The value of current $I_1$ is negative, So the current flow in the curcuit, is in opposite direction.

The current in 3Ω resistor is $I_1 + I_2$ = −0.92 + 4.55 = 10A

## 2.6.5 Work, Power and Energy

We are going to study about the work, power and energy their inter-relation



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